

Dual-Stream CNN-BiLSTM Model with Attention Layer for Automatic Modulation Classification

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Abstract—The Automatic Modulation Classification (AMC) approach is utilized to determine the nature of modulation. It is a significant job for intelligent receivers, essential components of future wireless communication technologies, including adaptive modulation and cognitive radio. Deep learning is often utilized for automatic modulation classification because of its superior performance and learning capability. A dual-stream CNN and BiLSTM architecture are used in this paper to classify seven different forms of digital modulation (8PSK, CPFSK, GFSK, QAM64, QPSK, BPSK, QAM16). Additionally, an attention layer is added to the model to enhance performance. Moreover, using dual streams rather than a single one improves the quality of the output. The proposed model is compared with earlier works. In comparison to the current research, it is observed that the proposed model provides superior classification accuracy with less complexity. The time needed for classification is also decreased due to the reduction in complexity.

Index Terms—Automatic Modulation Classification, CNN, BiLSTM, Attention layer

I. INTRODUCTION

Artificial intelligence applications are becoming increasingly prevalent in wireless communication due to improved technology, such as adaptive modulation and cognitive radio. The effectiveness of these technologies depend on the knowledge available on the radio spectrum, such as the frequency bands that are in use or not, the sort of modulation employed in the radio signal, etc. An essential function of an intelligent receiver is automatically detecting the modulation type of a radio signal at the receiver.

Blind modulation identification is difficult without knowledge of the sent data and unknown receiver characteristics, such as the phase offsets, signal amplitude, timing information, carrier frequency, etc. The task gets more complicated in a real-world situation with time-varying channels and multi path fading. It is challenging for communication engineers to build better information services and systems for military applications in a congested electromagnetic spectrum. In such situations, social signals must be safely delivered and received, while hostile signals must be tracked down, recognized, and blocked. In such circumstances, enhanced procedures are needed for real-time signal interception and processing, which is crucial for choices regarding electronic warfare and other critical tactical operations. Blind modulation format identification is a severe issue even with commercial devices like software-defined radio (SDR). Typically sent supplementary information is used to adjust the SDR system. When paired with an intelligent receiver, blind techniques can be utilized to increase transmission efficiency while minimizing overhead.

The two primary forms of modulation classifiers are likelihood-based and feature-based. Over the years, various

techniques, from conventional to machine learning techniques, have been attempted for automated modulation classification. Deep learning algorithms have become increasingly popular due to autonomous feature extraction and improved classification accuracy.

Likelihood-based (LB) algorithms for linear digital modulation classification are explored in paper [1]. In [2], the author proposes a fuzzy technique for multiclass categorization. Signals are retrieved with the help of four characteristics, including Renyi entropy, zero crossing rate, skewness, and frequency change. A support vector machine (SVM) is given these attributes to classify three digital modulations (4-ASK, 8-FSK, 8-PSK). At 20 dB SNR, this model can deliver an accuracy of around 77%, while at low SNR (5 dB), their performance is just 55%. In [3], the authors have proposed SVM based classification technique, where signal spectral correlation is employed as an SVM feature. Eleven modulation schemes, including analog and digital modulation schemes, are categorized using two layers of convolutional neural network (CNN) followed by two fully connected dense layers in [4]. A CNN model for AMC is proposed in [4] performs superior to traditional methods based on expert features. In [5], the author ignored the association of various features and trained a model (CLDNN) based on CNN and long short-term Memory (LSTM) utilizing just I/Q representation. A CNN model is also used to operate on the received signals' amplitude and phase data, and it observed a small increase in performance in [6]. In [7], a taxonomy of modulations based on constellation diagrams is presented. In [8], the authors have proposed a two combined CNN model. In [9], a feature image-based classification method is presented. In this method, feature values are taken from received signals, transformed into two-dimensional images, and provided as an input to CNN. The high influence effective feature is classified using the deep learning method proposed in [10]. In [11], a polar coordinate deep learning technique is used to minimize the impact of carrier phase shift. LSTM is introduced to the CNN model in paper [12] to learn time-related information in addition to spatial ones. The residual and inception structure is integrated with the LSTM module for the classification task. A Dense layer Dropout CNN (DDrCNN) based AMC model is proposed in [13]. In [14], the author proposed CNN-based zero-forcing (ZF) equalization AMC method for multiple input multiple output (MIMO) systems. A hybrid two-stream model for modulation classification is proposed in [15]. To optimize accuracy by extracting both spatial and temporal information, the CNN and LSTM layers model is

presented. In order to obtain high classification accuracy, an AMC technique based on deep learning is proposed in [16]. It combines random erasing with an attention mechanism. In [17], research on several deep learning architectures with sophisticated and enhanced designs for better performance in the presence of channel impairments is discussed.

In this work, we propose a dual stream CNN-BiLSTM model with attention layer for AMC. Since the digital modulation schemes are more popular than the analog modulation schemes, and the next generation of wireless communication mostly deals with digital modulation schemes, in this work, we have considered only digital modulations schemes. It is demonstrated using experiments that the proposed algorithm performs better than the existing techniques in terms of both classification accuracy and complexity.

II. SIGNAL MODEL

Using the standard wireless impulse response model, the received signal $y(t)$ is denoted by

$$y(t) = \exp[j \cdot \Delta f(t)] \int_{t=0}^{\tau} s[\Delta c(t - \tau)] h(t, \tau) d\tau + n(t), \quad (1)$$

where j is the imaginary number, τ is maximum delay spread, $s(t)$ is noise free complex transmitted signal, $h(t, \tau)$ is fading channel gain, $n(t)$ is additive white Gaussian noise (AWGN) with zero mean and σ_n^2 variance, $\Delta f(t)$ and $\Delta c(t)$ are carrier frequency and sampling rate offset respectively.

We get $y[n]$ from $y(t)$ with $f_s = 1/T_s$ sampling rate. $y[n]$ has both in-phase $y^I \in \mathbb{R}$ and quadrature components $y^Q \in \mathbb{R}$, and it can be represented as

$$y(n) = y^I[n] + jy^Q[n]. \quad (2)$$

Let us assume that N samples are obtained in a sampling period, then j^{th} data vector can be written as

$$y_j = [y_j[0], y_j[1], y_j[2], \dots, y_j[N-1]]^T, \quad (3)$$

where the T , superscript is the transpose operator.

Let $y_j \rightarrow X_j^{I/Q}$ denotes in-phase/quadrature I/Q vector and $y_j \rightarrow X_j^{A/P}$ denotes amplitude/phase A/P vector of $y_j \in \mathbb{C}^N$. For different data representations, the relevant transformations are as follows.

- 1) Transformation 1 (I/Q vector): Let the in-phase and quadrature elements of the I/Q vector be represented by X_j^I and X_j^Q , respectively. The dataset of raw complex samples serve as the source of the I/Q vector. The I/Q vector can be written as

$$X_j^{I/Q} = \begin{bmatrix} X_j^{I^T} \\ X_j^{Q^T} \end{bmatrix}, \quad (4)$$

where $X_j^{I^T}, X_j^{Q^T} \in \mathbb{R}^N$

- 2) Transformation 2 (A/P vector): The A/P vector consist of two vector elements, the amplitude X_j^A vector and the phase X_j^P vector can be represented as

$$X_j^{A/P} = \begin{bmatrix} X_j^{A^T} \\ X_j^{P^T} \end{bmatrix}, \quad (5)$$

where $X_j^{A^T}, X_j^{P^T} \in \mathbb{R}^N$.

The magnitude and phase vectors transformations can be written using Eq. (2) as

$$x_j^A[n] = \sqrt{y_j^I[n]^2 + y_j^Q[n]^2}, \quad (6)$$

$$x_j^P[n] = \arctan\left(\frac{y_j^Q[n]}{y_j^I[n]}\right). \quad (7)$$

III. PROPOSED ARCHITECTURE

The proposed model is a hybrid model of CNN, BiLSTM, and the attention layer as shown in Fig. 1. The CNN has superiority in extracting a spatial feature, and it can convolute signals and provide insightful high-level feature. Likewise, the BiLSTM is helpful for signal entity recognition and may learn gate weights by considering the past cell state and input data to solve the defined goal. Using attention allows configuring the model to perform recognition by focusing on a specific processed signal. The proposed architecture is implemented for classifying digital modulation schemes, i.e., 8PSK, CPFSK, GFSK, QAM64, QAM16, QPSK, and BPSK. We consider I/Q and A/P data as an input to the proposed model.

A CNN-BiLSTM with attention stream will process each component of the input. Since the structures of these two streams are similar, only Stream 1 structure is discussed in depth here for the sake of simplicity. In the proposed architecture of the model, one convolution layer is used in Stream 1 to extract spatial features from various representations of modulation signals. The BiLSTM layer is added after the convolution layer and is followed by the attention layer in order to make use of the temporal correlation.

The important task of convolution layers is the feature extraction and feature map by the convolution kernel. The i -th convolution layer uses 256 kernels, each of which has a size of 2×3 , and the rectified linear unit (ReLU) is adopted as the activation function. Additionally, zero-padding is mandated before each convolution layer to control the output form.

A 260×130 feature map is obtained after the convolution layer. It will be provided as input of the BiLSTM layer with 25 memory units. A 50×130 feature map is obtained from BiLSTM layer and provided as input to the next Attention layer. And 50×1 feature vector is obtained from this layer. In order to prevent overfitting, the dropout approach is further used on both convolution layer and BiLSTM layer with a dropout rate of $p = 0.5$. The process in Stream 2 is the same as in Stream 1.

An outer product is carried out to bring out feature diversity once the feature vectors from both streams have been obtained. Equation 8 represents the outer product mathematically. If f_{s1} and f_{s2} represent the feature functions of streams 1 and 2, respectively, with the appropriate input I/Q vector $X_j^{I/Q}$

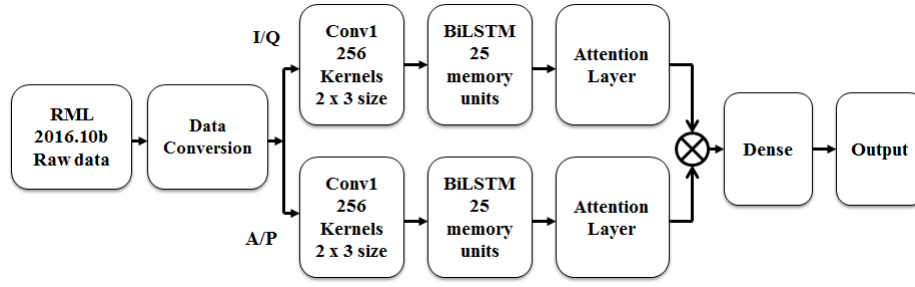


Fig. 1: CNN-BiLSTM with Attention layer proposed architecture.

and A/P vector $X_j^{A/P}$, the final features may be derived by applying the following equation

$$f_{outerproduct} = f_{s1} \left(X_j^{I/Q} \right) f_{s2} \left(X_j^{A/P} \right)^T. \quad (8)$$

A 50×50 shaped output after the outer product is reshaped using a flatten layer. The output of the flatten layer is 2500 dimensional feature vector. After that, the feature vector is merged using a dense softmax layer. This step converts the gathered characteristics into several modulation schemes.

The output of model represented as

$$\hat{Z} = F \left(X_j^{I/Q}, X_j^{A/P}, \alpha \right), \quad (9)$$

where $F(\cdot)$ is the overall function of classifier and α is the model parameter obtained by training.

We have considered the cross entropy loss function to produce high accuracy and quick convergence with less computation. The mathematical expression for the loss function is represented by

$$L(\alpha) = -\frac{1}{N} \sum_{k=1}^N z_k \log \hat{z}_k + (1 - z_k) \log (1 - \hat{z}_k), \quad (10)$$

where α is the parameter, N is the number of samples in training batch and z_k is the target output, $\hat{z}_k$ is the predicted output.

The Adam (adaptive moment estimation) optimizer is employed to maximize the loss function. Adam computes more quickly and needs fewer tuning settings. According to the E. 11, the network parameters are changed iteratively. .

$$\alpha_{t+1} = \alpha_t - \eta \Delta L(\alpha_t), \quad (11)$$

where η represents the learning rate, Δ is the gradient operator and $L(\alpha_t)$ represents the loss function.

The validation loss and α undergo value modifications with each iteration until the validation loss reaches the minimal feasible value. Training process is as follows

- 1) Model parameter α is first randomly initialized.
- 2) Training set X is randomly separated into mini -batches.
- 3) One mini-batch X_b is randomly choosen.
- 4) Conversion of X_b signal into I/Q and A/P format.
- 5) To obtain final predictive label distribution the model is utilized.
- 6) Error between true and predictive label distributions is calculated using Eq. (10).

- 7) Parameter α is updated on X_b by reverse propagation using Eq. (10) and Eq. (11).
- 8) Steps (4) - (7) is repeated until loss converges.

IV. EXPERIMENTAL DATASET

A. Dataset

RML2016.10B is used in this work. It is an open source dataset and can be downloaded from [18]. This dataset contains ten modulation schemes (8PSK, BPSK, CPFSK, PAM4, QAM16, QPSK, AM-DSB, WBFM, GFSK, QAM64), digital and analog modulation schemes. The dataset consist of 1200000 examples i.e. 6000 examples for each modulation type per SNR. Each example is in the form of inphase and quadrature components. Examples are available for the SNR varying from -20 dB to 18 dB with step size of 2 dB. Error effects such as random walk drifting of carrier frequency oscillator and sample time clocks, time varying multipath fading of the channel impulse response and AWGN are added in channel environment during the data acquisition. The dataset simulates real-time radio communication signals using different modulations in several SNRs. The detailed specifications of RML 2016.10b dataset is as shown in Table I. As this work is focused on classification of digital modulation schemes, so only digital modulations (8PSK, CPFSK, GFSK, QAM64, QPSK, BPSK, QAM16) from the dataset are considered.

B. Training and testing details

Raw data is divided into 80% training and 20% testing to carry out the operation. The epoch and batch size are set to 100 and 1024 correspondingly after several iterations. It employs the Adam optimizer with a 0.001 learning rate. Training and testing data are composed of around 768000 samples and 192000 samples, respectively. The training procedure is carried out on a Quadro RTX 6000 GPU with 24GB of RAM and version 11.2 of CUDA. The libraries used for implementation are keras and tensorflow.

V. RESULTS AND DISCUSSIONS

Fig 2 illustrates the model's accuracy fluctuation with epoch. It is clear from the results in Fig 2 that the proposed model does not exhibit overfitting, which was a concern with the earlier models. It shows that accuracy grows with the number of epochs and reaches saturation at 80 epochs. The plot displays the accuracy for both training and validation.

TABLE I: RadioML2016.10b Dataset Parameters.

Parameter	Value
Modulation schemes	8PSK, CPFSK, QAM16, GFSK, QAM64, PAM4, QPSK, AM-DSB, BPSK, WBFM
Number of examples per SNR for each modulation	6000
Length per example	128
Sampling frequency	1 MHz
Signal format	In-phase and Quadrature (IQ)
Signal dimension	2×128 per example
Duration per example	128 μ s
SNR range	[-20dB : 2dB : 18dB]
Total number of examples in dataset	1200000

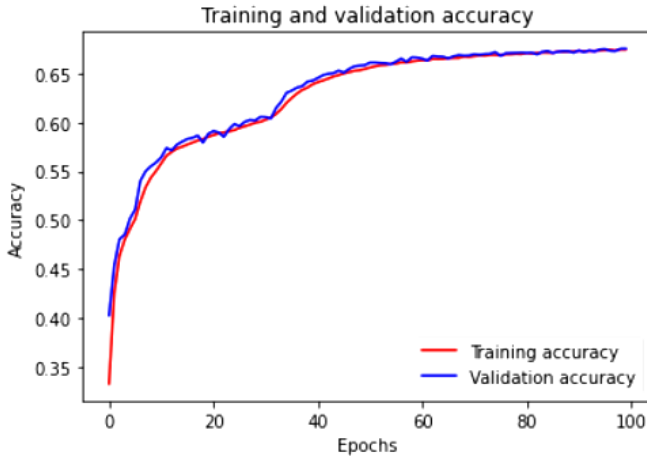


Fig. 2: Accuracy Vs epochs.

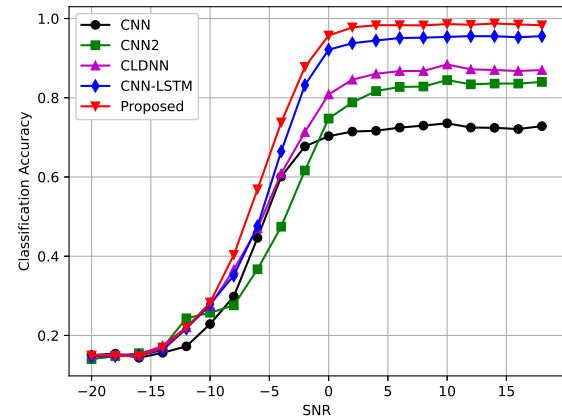


Fig. 3: Comparison of proposed model with deep learning models.

The accuracy versus SNR for the proposed model is provided in Fig. 3 for a more thorough analysis. From -4 dB to 18 dB, the proposed model provides more than 75 % accuracy. The accuracy is more than 96 %, from 2 dB to 18 dB. The proposed model's performance is better with a model from CNN-LSTM [15], CLDNN [5], CNN2 [6] and CNN [4] in terms of effectiveness. We trained these models with a batch size of 1024 and 100 epochs for digital modulation schemes and utilized the results for comparison with the proposed model. The model proposed in [15] has been determined to significantly outperform standard machine learning techniques such as including KNN, Linear SVM, Random Forest, Gaussian Naive Bayes, and expert SVM. We do not compare these machine learning methods in this research since it is clear that if the proposed model significantly outperforms the model in [15], it would also perform superior to these ML algorithms. In the SNR region of -16 dB to 20 dB, the proposed model's classification accuracy is noticeably more significant than the reference model's. These outcomes demonstrate that the proposed model is more effective at classifying digital modulation.

Fig. 4 depicts the confusion matrix for the proposed model at all SNR range. The confusion matrix provides a clearer picture of the classifier's performance. Fig. 5 - 7 display the confusion matrix for SNRs of ~ 6 dB, 0 dB, and 18 dB. The performance of the proposed model at low and high SNR is displayed using a graphic of the three distinct SNR

TABLE II: Comparison of Different Architecture Results.

Model	Trainable Parameters	Accuracy	Training Time (min-utes)	Testing Time (sec-onds)
Architecture in [15]	1149528	64.5	219	8
Proposed Model	145851	68.23	59	2.9

confusion matrices. Fig. 6 shows the proposed model's higher performance at 0 dB, where all of the confusion matrix's diagonal members are near to 1. It shows that the model can adequately identify the modulation schemes at SNR 0 dB. Fig. 5 illustrates how the model may provide more than 50% accuracy even at SNR $= -6$ dB. As seen in Fig. 7, all diagonal elements are equal to 1 at 18 dB SNR, demonstrating the model's improved performance at high SNR.

Table II compares the proposed model's accuracy, complexity, training time, and testing time to the model in [15]. The model in [15] is provided with the same dataset that is used as input for the proposed model in this study to compare the outcomes. The table II shows that the proposed model produces better outcomes than model in [15]. The proposed model decreases model complexity and, as a result classification time, and increases classification accuracy.

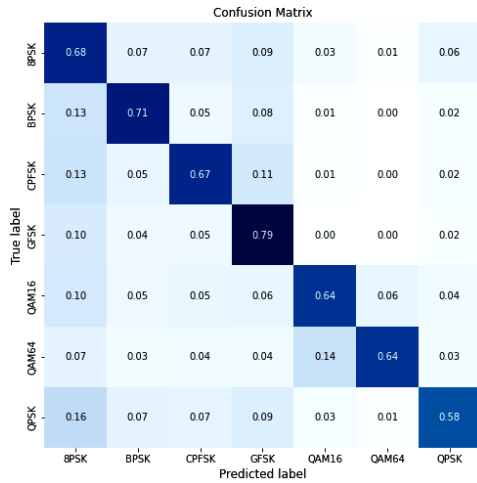


Fig. 4: Confusion matrix of proposed model for all SNR values.

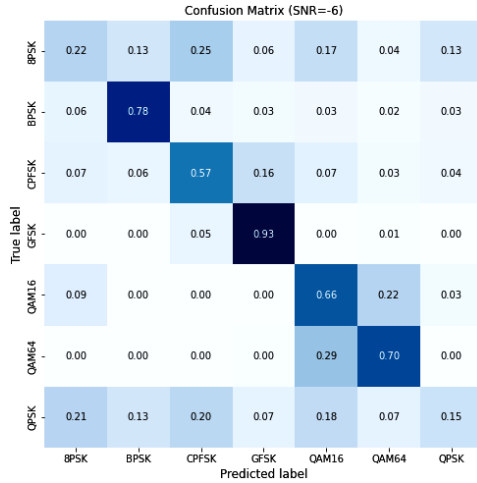


Fig. 5: Confusion matrix of proposed model for -6 dB SNR.

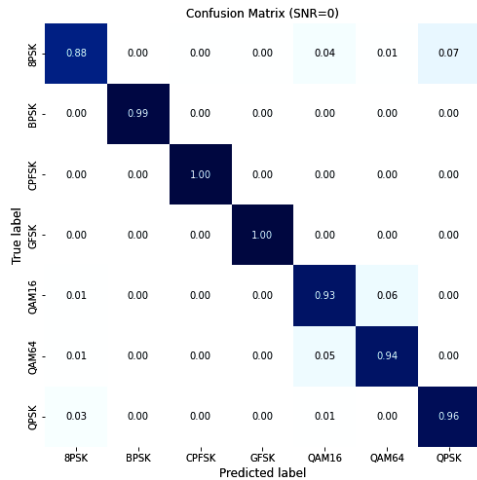


Fig. 6: Confusion matrix of proposed model for 0 dB SNR.

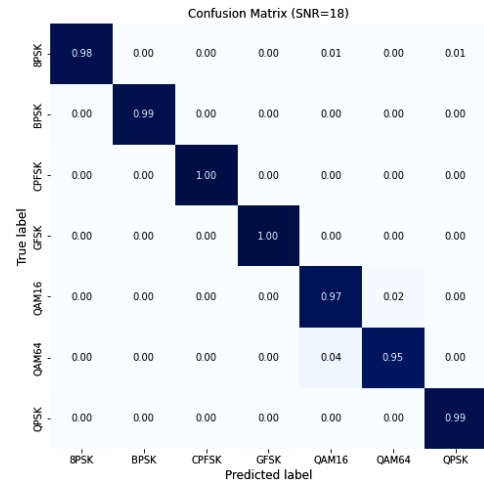


Fig. 7: Confusion matrix of proposed model for 18 dB SNR.

VI. CONCLUSION

This paper proposes an automatic modulation classification using a dual stream CNN-BiLSTM with attention layer architecture. Combining CNN and BiLSTM improves the learning of spatial and temporal information. The attention layer aids in the memorization of lengthy data sequences. This model works best for categorizing digital modulation techniques with amplitude, phase, and I/Q as input, for a broad range of SNR. It is clear from the simulation results that the proposed classifier can effectively learn and categorize the radio signal. The complexity and computing time of the classifier is reduced with fewer layers and filters, allowing the classifier to be used in real-life situations.

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